



Q1 Sum the given number upto single digit

5 marks

Write a Python code to find the sum of the given number upto single digit

input: 89789

output : 5

Your Code

```
1 n = input("")
2 while len(n)>1:
3     total=0
4     for i in n :
5         total = total + int(i)
6         n=str(total)
7 print(n)
```

Input

89789

Output

5



Q2 Prime Number

5 marks

Find the next prime number

input: 13

output: 17

Your Code

```
1 n =int(input(" "))
2 n = n+1
3 while True:
4     count=0
5     for i in range(1,n+1):
6         if(n%i ==0):
7             count=count+1
8     if (count==2):
9         print(n)
10        break
11    n=n+1
```

Input

13

Output

17



Q3 Swap the first and last digits

5 marks

WAP to swap the first and last digits of the given number.

Input: 23546

Output: 63542

Your Code

```
1 a= (input(""))
2 n= str(a)
3 if len(n)==1:
4     print("swapped number",n)
5 else:
6     n= n[-1]+n[1:-1]+n[0]
7     print(n)
```

Input

23456

Output

63452



Q4 Odd Sequence first and Even Sequence next

5 marks

WAP to odd Sequence first and Even Sequence next

Input: 658134

Output: 513684

Your Code

```
1 a =input("")
2 odd=""
3 even=""
4 for i in a:
5     if int(i) %2==0:
6         even=even+i
7     else:
8         odd=odd+i
9 print(odd+even)
```

Input

658134

Output

513684

SIMATS Online Programming Lab

PYTHON PROGRAMMING



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54567 → Yes

|5-4| = 1

|4-5| = 1

|5-6| = 1

|6-7| = 1

1235 → No

|3-5| = 2

Your Code

```
1 num = input("")
2 s = str(num)
3 step = True
4 for i in range(len(s)-1):
5     if abs(int(s[i])-int(s[i+1])) != 1:
6         step = False
7         break
8 if step:
9     print("yes")
10 else:
11     print("no")
```

Input

54567

Output

yes



Q1 Reverse Only the even numbers

5 Marks

WAP to reverse only the even numbers

Input: 325649

Output: 345629

Your Code

```
1 num =input("")
2 even = []
3 for ch in num:
4     if int(ch)% 2 ==0:
5         even.append(ch)
6 even.reverse()
7 result = ""
8 j = 0
9 for ch in num:
10     if int(ch)%2==0:
11         result +=even[j]
12         j +=1
13     else:
14         result += ch
15 print(result)
```

Input

325649

Output

345629



Q7 Encoding

5 marks

WAP to encode the given number

Input: 55553322255

Output: 54322352

Your Code

```
1 num = input("")
2 count = 1
3 for i in range(len(num)-1):
4     if num[i] == num[i+1]:
5         count += 1
6     else:
7         print(num[i]+str(count),end = "")
8         count = 1
9 print(num[-1]+str(count))
```

Input

55553322255

Output

54322352



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Q8 Reverse Odd Digit

5 marks

WAP to Reverse Odd Digits Only

Input: 816253

Output: 351

Your Code

```
1 num = input("")
2 for i in num[ : :-1]:
3     if int(i) %2 != 0:
4         print(i,end="")
```

Input

816253

Output

351



Q9 Convert every number into the next sequence number.

5 marks

WAP to convert every number into the next sequence number.

Input: 89781

Output: 90892

Your Code

```
1 num = input("")
2 for i in num:
3     if i == '9':
4         print(0,end="")
5     else:
6         print(int(i)+1,end="")
```

Input

89871

Output

90982



Q10 Print the total factor count of a given number

5 marks

WAP to print the total factor count of a given number

Input: 12

Output: 6

Your Code

```
1 n=int(input(""))
2 count=0
3 for i in range(1,n+1,1):
4     if(n%i==0):
5         count = count+1
6 print(count)
```

Input

12

Output

6